

UNIT 2: ENZYMES

Biology 9

Mr. Queenan

Text - Ch 2.4 (p 58-63)

Questions

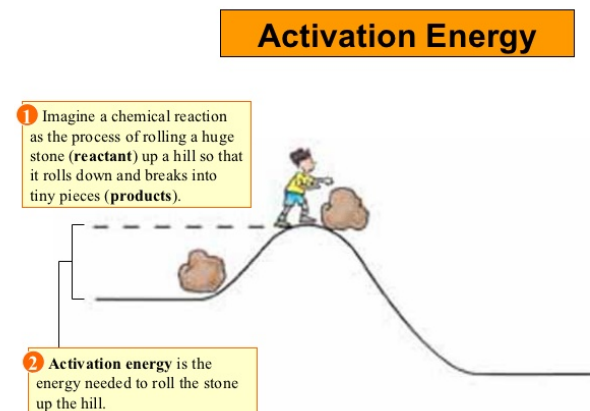
- You are lost in the woods, and nightfall is approaching. You need to make a fire.
- ▣ What materials do you need for the fire
- ▣ If you had access, what could be used to create a larger fire, or help you make the fire faster?

Enzymes

- Although biological molecules possess much potential energy, it is not released spontaneously.
 - ▣ An energy barrier must be overcome before a chemical reaction can begin.
 - ▣ This energy is called the **activation energy** (because it activates the reactants).

Enzymes

- We can think of activation energy as the amount of energy needed for a reactant molecule to move “uphill” to a higher-energy but an unstable state so that the “downhill” part of the reaction can begin.
- One way to speed up a reaction is to add heat, which agitates atoms so that bonds break more easily and reactions can proceed, but too much heat will kill a cell.

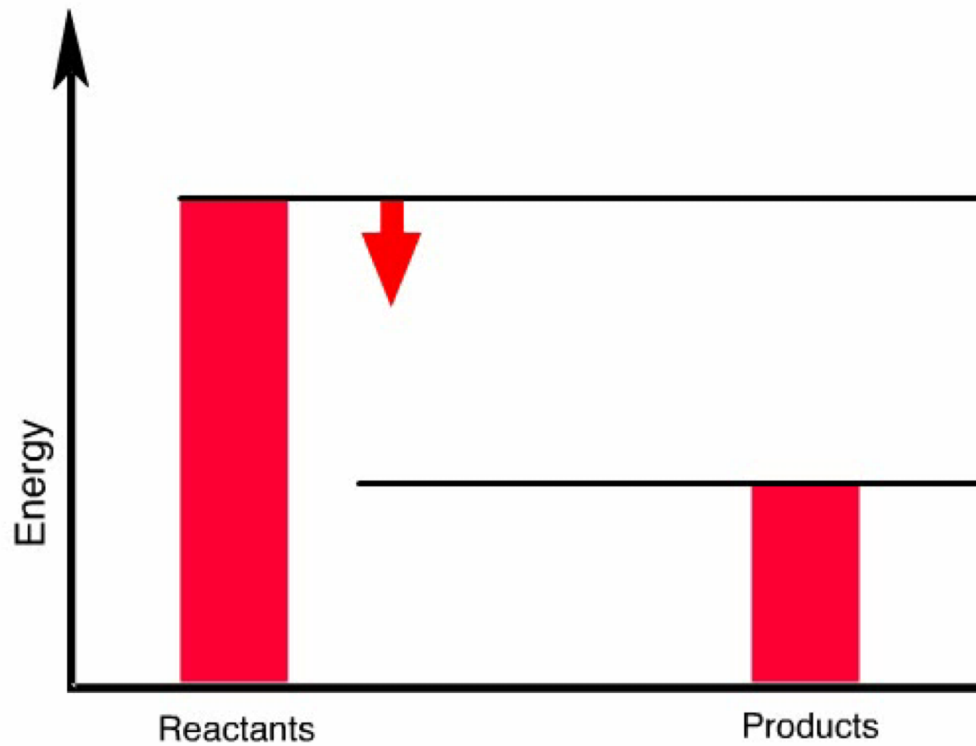


Enzymes

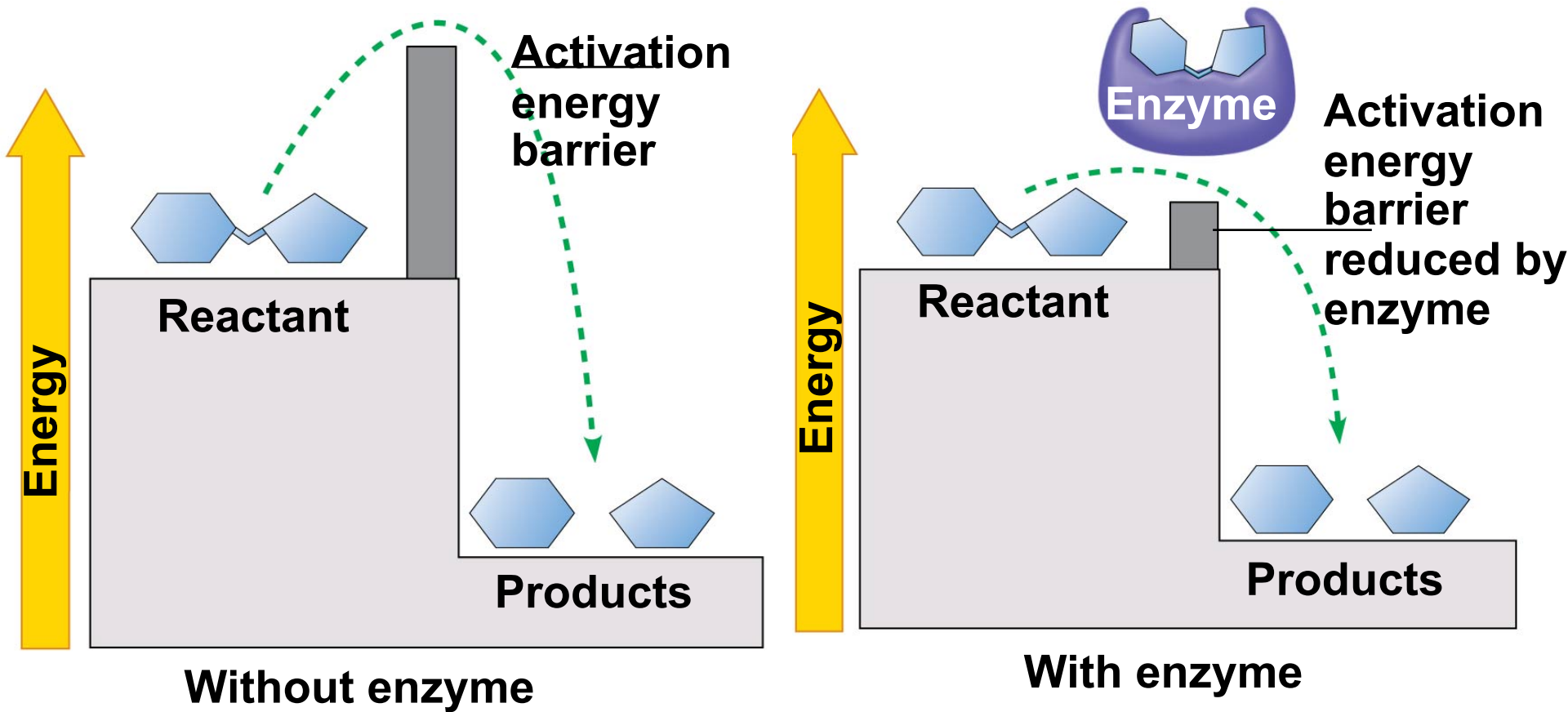
□ Enzymes

- ▣ function as biological catalysts,
 - ▣ increase the rate of a reaction without being consumed by the reaction, and
 - ▣ are usually proteins (although some RNA molecules can function as enzymes (**Ribozymes**)).
- Enzymes speed up a reaction by lowering the activation energy needed for a reaction to begin.

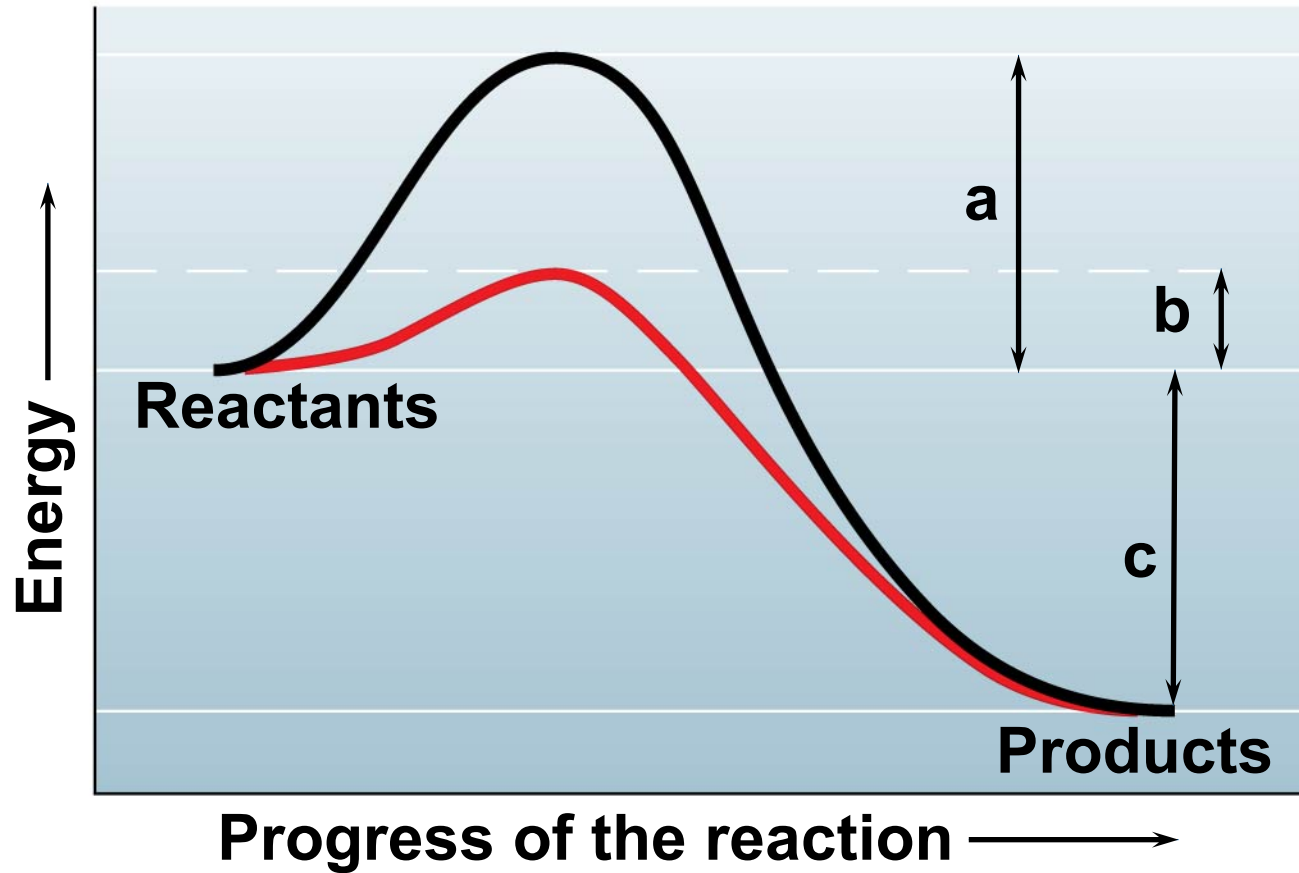
Enzymes



Enzymes



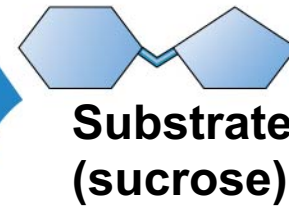
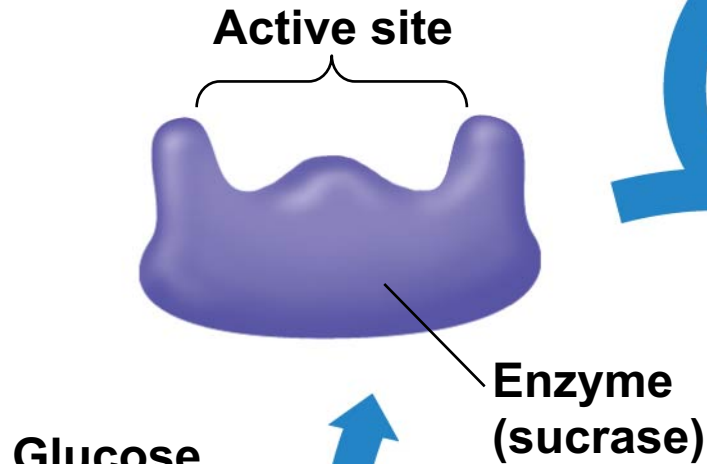
Enzymes



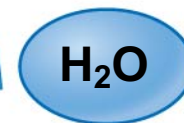
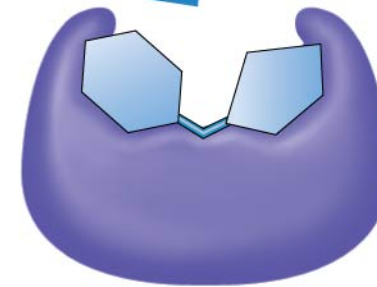
Enzymes

- An enzyme
 - is very selective in the reaction it catalyzes and
 - has a shape that determines the enzyme's specificity.
- The specific reactant that an enzyme acts on is called the enzyme's **substrate**.
- A substrate fits into a region of the enzyme called the **active site**.
- Enzymes are specific because only specific substrate molecules fit into their active site.

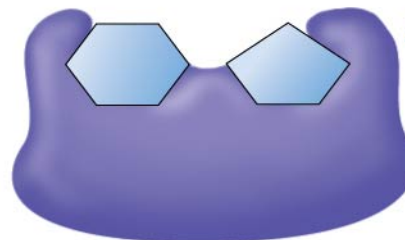
1 The enzyme available with an empty active site



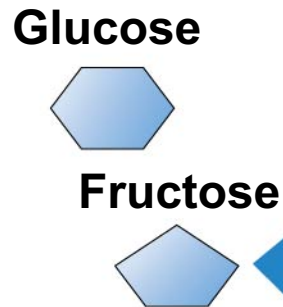
2 Substrate binds to enzyme with induced fit.



3 The substrate is converted to products



4 The products are released



Enzymes

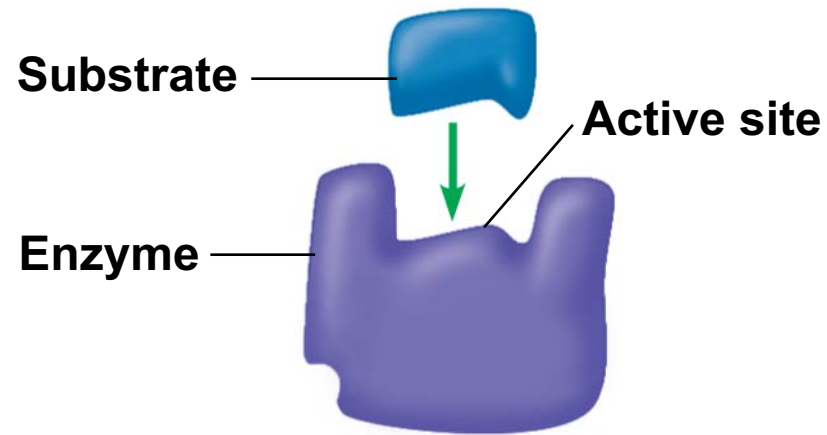
- For every enzyme, there are optimal conditions under which it is most effective.
- Temperature affects molecular motion.
 - An enzyme's optimal temperature produces the highest rate of contact between the reactants and the enzyme's active site.
 - Most human enzymes work best at 35–40°C.
- The optimal pH for most enzymes is near neutrality.

Enzymes

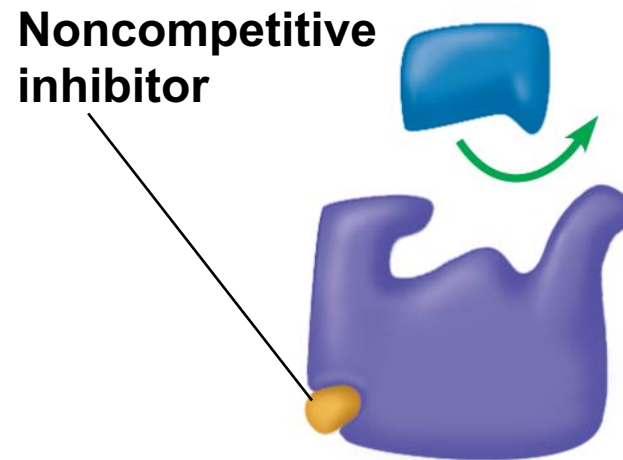
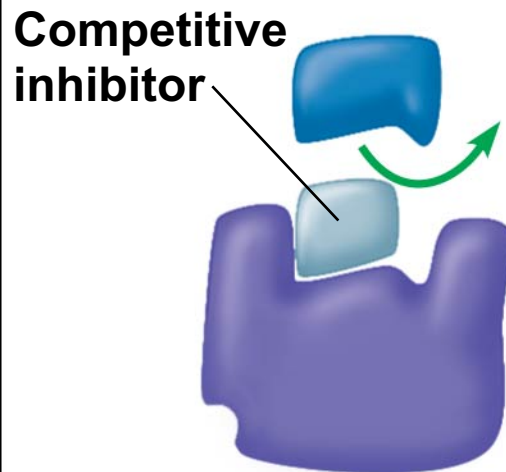
- Many enzymes require nonprotein helpers called **cofactors**, which
 - ▣ bind to the active site and
 - ▣ function in catalysis.
- Some cofactors are inorganic, such as the ions of zinc, iron, or copper.
- If a cofactor is an organic molecule, such as most vitamins, it is called a **coenzyme**.
 - ▣ Acetyl CoA, CoQ10
- ▣ Enzymes and coenzymes/cofactors work together to signal processes to start, run, and end throughout biological systems

Enzymes

- A chemical that interferes with an enzyme's activity is called an inhibitor.
- **Competitive inhibitors**
 - ▣ block substrates from entering the active site and
 - ▣ reduce an enzyme's productivity.
- **Noncompetitive inhibitors**
 - ▣ bind to the enzyme somewhere other than the active site,
 - ▣ change the shape of the active site, and
 - ▣ prevent the substrate from binding.



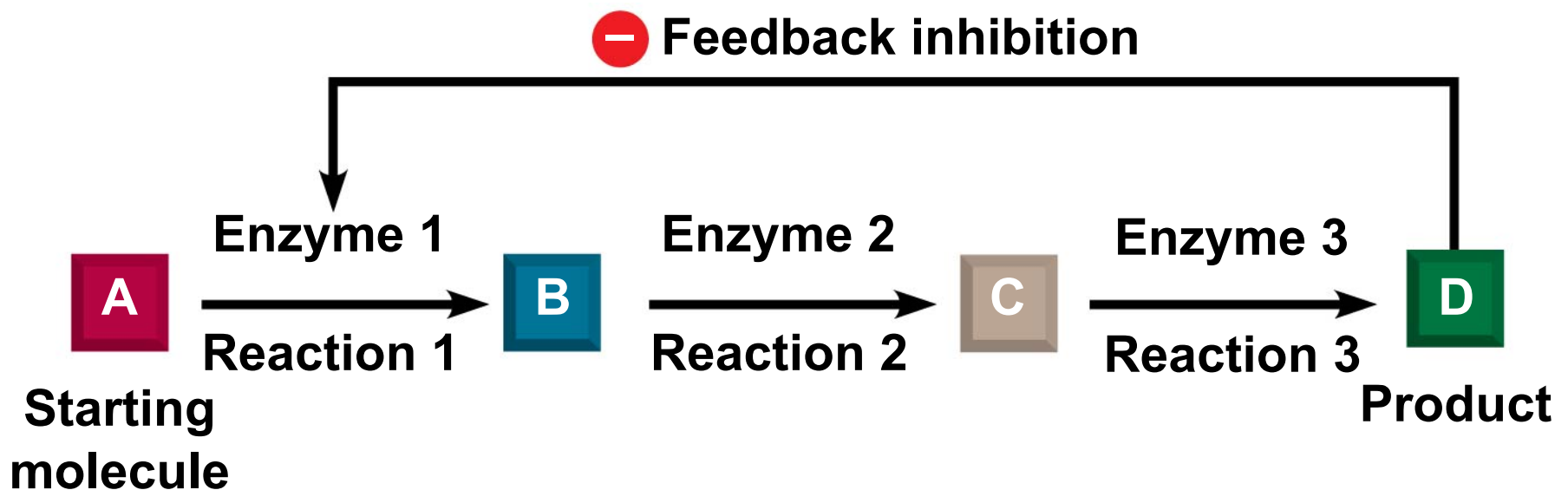
Normal binding of substrate



Enzyme inhibition

Enzymes

- Enzyme inhibitors are important in regulating cell metabolism.
- In some reactions, the product may act as an inhibitor of one of the enzymes in the pathway that produced it. This is called **feedback inhibition**.



Enzymes

- Many beneficial drugs act as enzyme inhibitors, including
 - ▣ ibuprofen, which inhibits an enzyme involved in the production of prostaglandins (messenger molecules that increase the sensation of pain and inflammation),
 - ▣ some blood pressure medicines,
 - ▣ some antidepressants,
 - ▣ many antibiotics, and
 - ▣ protease inhibitors used to fight HIV.
- Enzyme inhibitors have also been developed as
 - ▣ pesticides and
 - ▣ deadly poisons for chemical warfare.